Decorative Grille D-SS1/D-SS2





Description
Supply/exhaust aluminum grille with fixed blades in 1 or 2 directions **D-SS1/ D-SS2**.

Application
Supply/ exhaust of the air flow

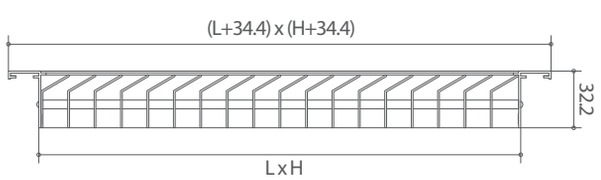
Characteristics
Fixed blades in 1 or 2 directions mounted parallel to length "L"
The grille is provided with mounting holes (standard).
The air flow can be adjusted by mounting an air control damper **OBD** on the grille inlet.

Accessories
Settlement frame made of galvanized sheet metal **CC**
Attachment spring **A**
Air control damper with opposable blades **OBD**
Air filter **FIL**
Insulated/Non-insulated adapter **ADP IZ/N**

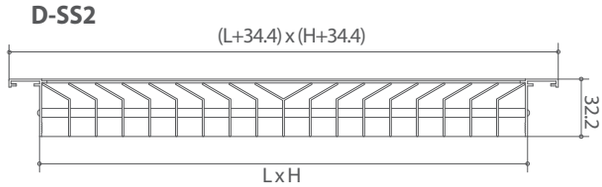
Materials and finishing
Extruded aluminum material, electrostatic field painted white RAL 9016
Other RAL colors available on request
Natural anodised aluminum

Technical design

D-SS1



D-SS2



Assembly Methods
Screws mounting (standard)
Settlement frame (optional)
Attachment spring (optional)

Selection table D-SS1 / D-SS2

L/H>2.1

Air flow (m^3/h)	Dimensions Ak [m^2]	500x200 0.05	600x200 0.06	800x200 0.08	1000x200 0.10	1200x200 800x300 0.12	1000x300 0.15	1200x300 1000x400 0.20	1200x400 0.25	1200x500 0.30
200	X [m] Veff [m/s] NR [dB(A)] ΔPt [pa]	<3 1.1 <20 <4	<3 0.9 <20 <4							
250	X [m] Veff [m/s] NR [dB(A)] ΔPt [pa]	3.0 1.4 <20 <4	<3 1.2 <20 <4	<3 0.9 <20 <4						
300	X [m] Veff [m/s] NR [dB(A)] ΔPt [pa]	3.5 1.7 <20 <4	3.3 1.4 <20 <4	3.0 1.0 <20 <4	<3 0.8 <20 <4					
350	X [m] Veff [m/s] NR [dB(A)] ΔPt [pa]	4.5 1.9 <20 <4	4.0 1.6 <20 <4	3.8 1.2 <20 <4	3.0 1.0 <20 <4	<3 0.8 <20 <4				
400	X [m] Veff [m/s] NR [dB(A)] ΔPt [pa]	5.0 2.2 <20 4.0	4.5 1.9 <20 <4	4.0 1.4 <20 <4	3.5 1.1 <20 <4	3.0 0.9 <20 <4				
450	X [m] Veff [m/s] NR [dB(A)] ΔPt [pa]		5.0 2.1 <20 <4	4.0 1.6 <20 <4	3.5 1.3 <20 <4	3.0 1.0 <20 <4				
500	X [m] Veff [m/s] NR [dB(A)] ΔPt [pa]		5.5 2.3 <20 4.5	5.0 1.7 <20 <4	4.5 1.4 <20 <4	4.0 1.2 <20 <4				
600	X [m] Veff [m/s] NR [dB(A)] ΔPt [pa]		6.5 2.8 20.0 6.0	6.0 2.1 <20 <4	5.3 1.7 <20 <4	5.0 1.4 <20 <4				
700	X [m] Veff [m/s] NR [dB(A)] ΔPt [pa]		9.0 3.2 25.0 8.5	8.0 2.4 <20 4.5	7.0 1.9 <20 <4	6.3 1.6 <20 <4				
800	X [m] Veff [m/s] NR [dB(A)] ΔPt [pa]			8.0 2.8 20.0 6.0	7.5 2.2 <20 4.0	7.0 1.9 <20 <4				
900	X [m] Veff [m/s] NR [dB(A)] ΔPt [pa]			9.5 3.1 23.0 8.0	8.5 2.5 <20 5.0	8.0 2.1 <20 <4				
1000	X [m] Veff [m/s] NR [dB(A)] ΔPt [pa]				9.0 2.8 20.0 6.0	8.0 2.3 <20 4.5				
1200	X [m] Veff [m/s] NR [dB(A)] ΔPt [pa]				12.5 3.3 28.0 9.0	11.0 2.8 20.0 6.0				
1400	X [m] Veff [m/s] NR [dB(A)] ΔPt [pa]				14.0 3.9 28.0 12.0	13.0 3.2 25.0 8.5	12.0 2.6 <20 5.5			
1600	X [m] Veff [m/s] NR [dB(A)] ΔPt [pa]					15.0 3.7 25.0 10.0	13.5 3.0 22.0 7.0	12.0 2.2 <20 4.0	11.0 1.8 <20 <4	
1800	X [m] Veff [m/s] NR [dB(A)] ΔPt [pa]					>15 4.2 30.0 14.0	15.0 3.3 28.0 9.0	13.0 2.5 <20 5.0	11.5 2.0 <20 <4	
2000	X [m] Veff [m/s] NR [dB(A)] ΔPt [pa]					>15 4.6 32.0 16.0	>15 3.7 25.0 10.0	14.0 2.8 20.0 6.0	12.5 2.2 <20 4.0	
2500	X [m] Veff [m/s] NR [dB(A)] ΔPt [pa]					>15 5.8 37.0 25.0	>15 4.6 32.0 16.0	>15 3.5 25.0 9.5	>15 2.8 20.0 6.0	15.0 2.3 <20 4.5
3000	X [m] Veff [m/s] NR [dB(A)] ΔPt [pa]					>15 6.9 43.0 38.0	>15 5.6 33.0 20.0	>15 4.2 30.0 14.0	>15 3.3 28.0 9.0	>15 2.8 20.0 6.0
3500	X [m] Veff [m/s] NR [dB(A)] ΔPt [pa]					>15 8.1 47.0 50.0	>15 6.5 40.0 30.0	>15 4.9 32.0 18.0	>15 3.9 28.0 12.0	>15 3.2 25.0 8.5
3600	X [m] Veff [m/s] NR [dB(A)] ΔPt [pa]							>15 5.0 37.0 24.0	>15 4.0 28.0 13.0	>15 3.3 28.0 9.0

Selection table D-SS1 , D-SS2

L/H<2.1

Air flow (m^3/h)	Dimensions	400x200	500x300	600x300	600x400	800x400
	Ak [m^2]	0.04	0.07	0.08	0.12	0.15
200	X [m] Veff [m/s] NR [dB(A)] ΔPt [pa]	<1 1.4 <20 <4	<1 0.8 <20 <4			
250	X [m] Veff [m/s] NR [dB(A)] ΔPt [pa]	<1 1.7 <20 <4	<1 1.0 <20 <4	<1 0.9 <20 <4		
300	X [m] Veff [m/s] NR [dB(A)] ΔPt [pa]	1.0 2.1 <20 <4	<1 1.2 <20 <4	<1 1.0 <20 <4	<1 0.7 <20 <4	
350	X [m] Veff [m/s] NR [dB(A)] ΔPt [pa]	1.2 2.4 <20 4.5	<1 1.4 <20 <4	<1 1.2 <20 <4	<1 0.8 <20 <4	<1 0.6 <20 <4
400	X [m] Veff [m/s] NR [dB(A)] ΔPt [pa]	1.4 2.8 20.0 6.0	1.0 1.6 <20 <4	<1 1.4 <20 <4	<1 0.9 <20 <4	<1 0.7 <20 <4
450	X [m] Veff [m/s] NR [dB(A)] ΔPt [pa]	1.6 3.1 23.0 8.0	1.2 1.8 <20 <4	1.0 1.6 <20 <4	<1 1.0 <20 <4	<1 0.8 <20 <4
500	X [m] Veff [m/s] NR [dB(A)] ΔPt [pa]	1.8 3.5 25.0 9.5	1.4 2.0 <20 <4	1.2 1.7 <20 <4	<1 1.2 <20 <4	<1 0.9 <20 <4
600	X [m] Veff [m/s] NR [dB(A)] ΔPt [pa]		1.8 2.4 <20 4.5	1.5 2.1 <20 <4	1.2 1.4 <20 <4	<1 1.1 <20 <4
700	X [m] Veff [m/s] NR [dB(A)] ΔPt [pa]		1.9 2.8 20.0 6.0	1.8 2.4 <20 4.5	1.5 1.6 <20 <4	1.2 1.3 <20 <4
800	X [m] Veff [m/s] NR [dB(A)] ΔPt [pa]		2.1 3.2 25.0 8.5	2.0 2.8 20.0 6.0	1.8 1.9 <20 <4	1.4 1.5 <20 <4
900	X [m] Veff [m/s] NR [dB(A)] ΔPt [pa]			2.2 3.1 23.0 8.0	1.9 2.1 <20 <4	1.5 1.7 <20 <4
1000	X [m] Veff [m/s] NR [dB(A)] ΔPt [pa]			2.5 3.5 25.0 9.5	1.9 2.3 <20 4.5	1.8 1.9 <20 <4
1200	X [m] Veff [m/s] NR [dB(A)] ΔPt [pa]			2.9 4.2 30.0 14.0	2.4 2.8 20.0 6.0	2.1 2.2 <20 4.0
1400	X [m] Veff [m/s] NR [dB(A)] ΔPt [pa]				2.8 3.2 25.0 8.5	2.4 2.6 <20 5.5
1600	X [m] Veff [m/s] NR [dB(A)] ΔPt [pa]				3.2 3.7 25.0 10.0	2.8 3.0 22.0 7.0
1800	X [m] Veff [m/s] NR [dB(A)] ΔPt [pa]				3.6 4.2 30.0 14.0	3.1 3.3 28.0 9.0
2000	X [m] Veff [m/s] NR [dB(A)] ΔPt [pa]				4.0 4.6 32.0 16.0	3.5 3.7 25.0 10.0
2500	X [m] Veff [m/s] NR [dB(A)] ΔPt [pa]					4.3 4.6 32.0 16.0
3000	X [m] Veff [m/s] NR [dB(A)] ΔPt [pa]					5.0 5.6 33.0 20.0
3500	X [m] Veff [m/s] NR [dB(A)] ΔPt [pa]					6.5 6.5 40.0 30.0
3600	X [m] Veff [m/s] NR [dB(A)] ΔPt [pa]					7.0 6.7 42.0 33.0

Annotation

* Ak [m²] - Free surface of the grille
* Veff[m/s] - Air velocity in grille
* ΔPt [pa] - Pressure loss
* X [m] - Air jet length
* NR [dB(A)] - Noise level without the room attenuation
* Vk = 0.5 m/s - Speed at which airflow length was calculated "x" [m]

Order codes

Type + **CC** + **A** (Attachment spring) + **FIL** + **OBD** + **ADP IZ/N** / **Dimensions** / **O**

- **Type**: **D-SS1 / D-SS2**
- **CC**: Settlement frame
- **A** (Attachment spring): No. of springs depending on grille dimensions
- **FIL**: Air filter
- **OBD**: Air control damper
- **ADP IZ/N**: Insulated/ Non-insulated adapter
- **Dimensions**: On request
- **O**: **O** Painted white RAL 9016; **R** Other RAL paints; Natural anodised